

Explainable Bitter Gourd Freshness Detection using MobileNetV3 and GradCAM

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MOTIVATION

- >50% of fruit and vegetable yield is lost along the global supply chain, much of it from spoilage that goes unnoticed.
- Manual inspection is slow, subjective, and inconsistent at scale.
- Bitter gourd is a high-value medicinal crop, yet automated freshness assessment for it remains *underexplored*.
- Deep models work, but their *black-box* nature is a barrier in food-safety contexts.

Goal: a lightweight, mobile-deployable, *explainable* classifier for fresh vs. rotten bitter gourd.

CONTRIBUTIONS & HIGHLIGHTS

- Explainable freshness classifier** for bitter gourd combining MobileNetV3-Small (transfer learning) with GradCAM visual explanations.
- Mobile/edge-ready:** lightweight backbone designed for low-resource deployment in post-harvest field settings.
- Balanced pipeline:** stratified 80/20 split, ImageNet-normalised inputs, and targeted augmentation (flip, rotation, jitter, grayscale).
- Perfect test performance:** 100% accuracy, precision, recall and F1 on a held-out test set of 40 images (20 fresh, 20 rotten).
- Trustworthy decisions:** GradCAM confirms that predictions are driven by *semantically meaningful fruit regions*, not background or spurious cues.

PROPOSED SYSTEM

The pipeline has four stages; (1) data preparation, (2) MobileNetV3 transfer-learning, (3) evaluation, and (4) GradCAM explainability, producing both a class label *and* a visual rationale for every prediction.

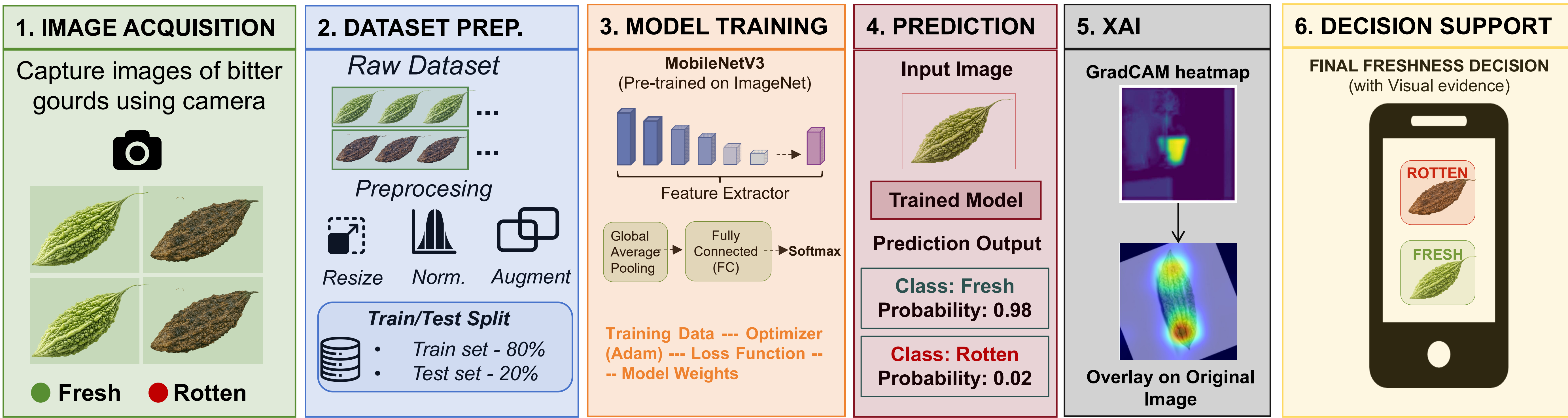


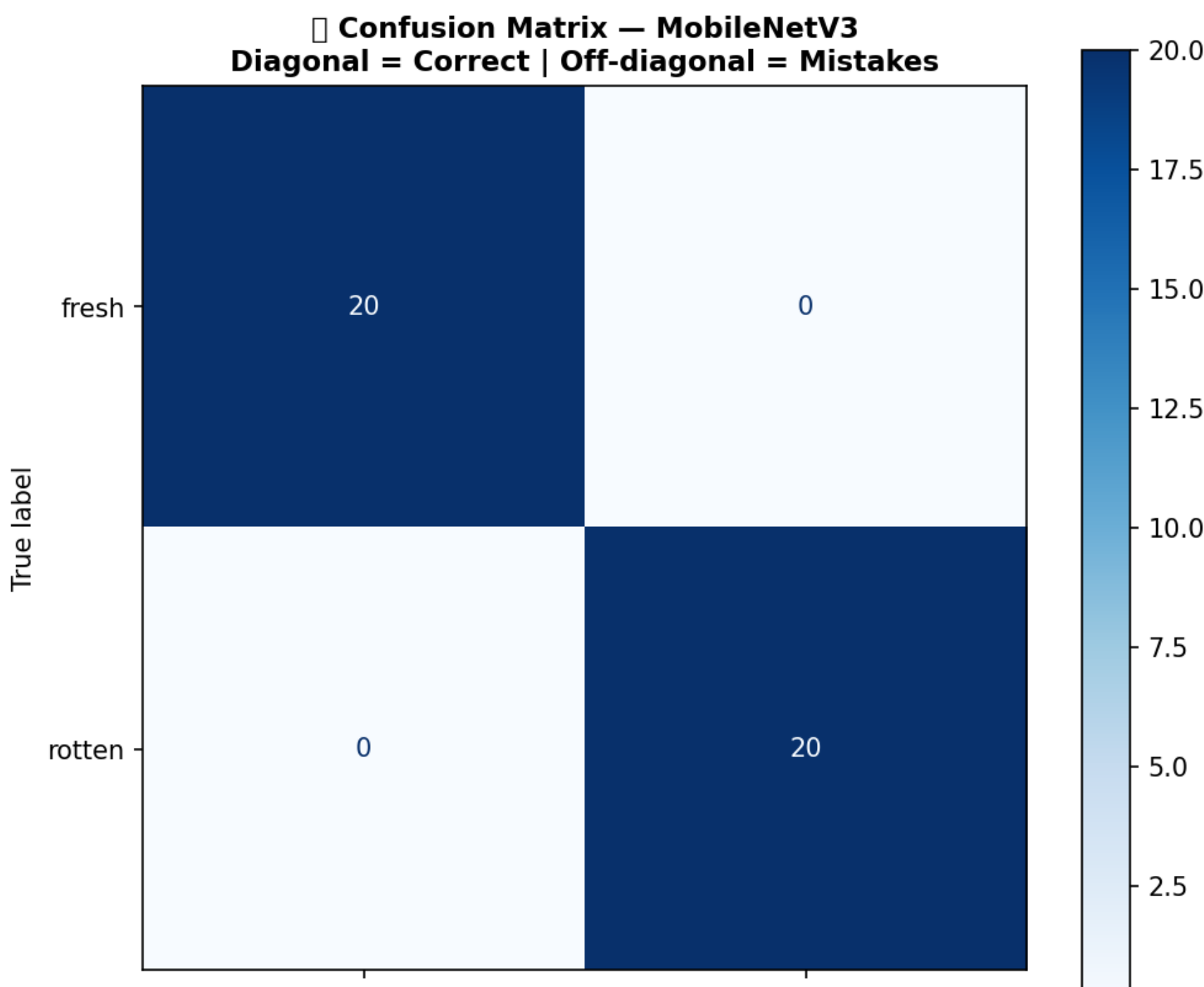
Fig. 1: End-to-end pipeline; preprocessing, MobileNetV3 classifier, and GradCAM explanation layer.

Backbone	MobileNetV3-Small, pretrained on ImageNet-1K; final FC replaced with Linear($\cdot \rightarrow 2$)
Input	RGB images resized to 224×224 , normalised with ImageNet mean/std
Augmentation	Horizontal flip ($p=0.5$), rotation $\pm 15^\circ$, colour jitter, grayscale ($p=0.1$)
Optimiser	Adam, $\text{lr}=1 \times 10^{-4}$, weight decay 1×10^{-4} , CrossEntropyLoss
Schedule	ReduceLROnPlateau (patience 3, factor 0.1) over 10 epochs, batch size 32
Hardware	Google Colab T4 GPU
Explainability	GradCAM on the final convolutional block of MobileNetV3

QUANTITATIVE RESULTS

On a held-out test set of **40 images** (20 fresh, 20 rotten):

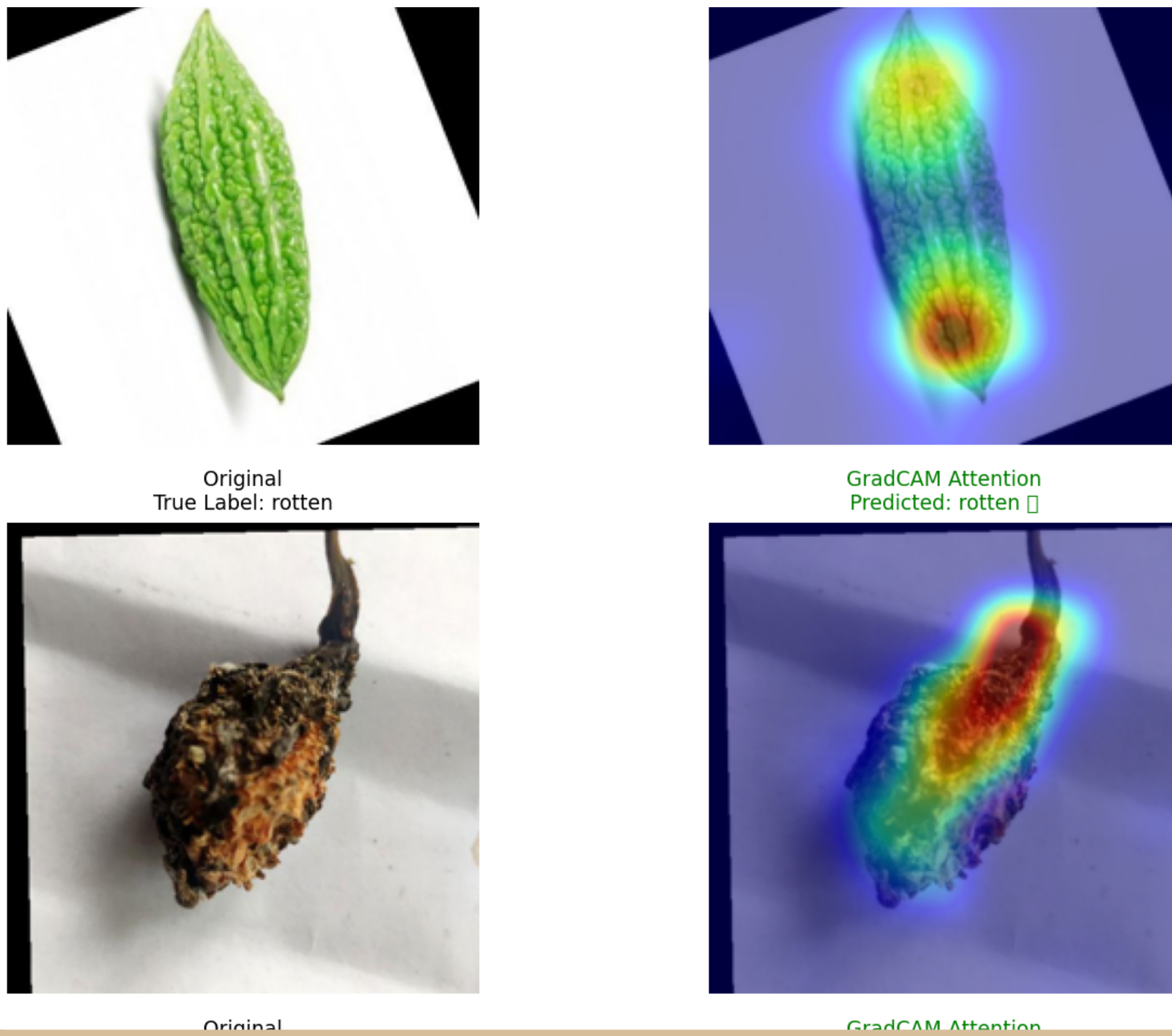
Class	Prec.	Rec.	F1	Sup.
Fresh	1.00	1.00	1.00	20
Rotten	1.00	1.00	1.00	20
Overall accuracy	100%			



GRADCAM EXPLAINABILITY

GradCAM produces class-activation heatmaps on the final convolutional block, revealing **where** the model looks when deciding “fresh” vs “rotten”.

- Fresh:** attention on the healthy central fruit body.
- Rotten:** attention on decayed stem & deteriorated tissue.
- Decisions are anchored in **semantically meaningful** regions, not background.



WHY IT MATTERS

- Lightweight + accurate:** MobileNetV3 is small enough for phones yet hits 100% on the test set.
- Transparent:** GradCAM turns predictions into visual evidence operators can trust.
- Practical impact:** faster, more consistent post-harvest sorting $\Rightarrow$ less food waste, safer supply chains, better margins for growers.
- Generalisable approach:** the explainable transfer-learning recipe extends to other under-explored crops.

100% accuracy with built-in explanations, ready for mobile post-harvest deployment.

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